

VMware

vSAN

STUDY NOTES

# VMware vSAN

## Comprehensive Study Notes

*For VCP-DCV · VCAP-DCV · VMware vSAN Specialist Certification*

### **Covers All Exam Domains:**

Architecture · Storage Policies · Networking · Security · Troubleshooting · ESA · Stretched Cluster

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## CHAPTER 01 · VSAN FUNDAMENTALS & ARCHITECTURE

### What is VMware vSAN?

VMware vSAN (Virtual SAN) is a **software-defined storage (SDS)** solution natively integrated into VMware ESXi. It aggregates locally attached storage devices across multiple ESXi hosts in a cluster to create a single, shared datastore — eliminating dependency on external SAN or NAS arrays.

#### ▣ TIP: Core Value Proposition

vSAN delivers enterprise-grade shared storage using commodity server-local drives, reducing TCO by 40–60% versus traditional SAN in many workloads.

### vSAN Architecture Layers

#### Layer 1 — VM Storage Policy

Policy-driven storage: each VM/VMDK is assigned a VM Storage Policy that governs how vSAN places, protects, and serves its data.

#### Layer 2 — Distributed Object Manager (DOM)

DOM manages distributed storage objects. It coordinates I/O, replication, and component placement across all hosts. Every host has a DOM Client (issues I/O) and a DOM Owner (coordinates writes to all replicas).

#### Layer 3 — Cluster Monitoring Membership & Directory (CMMDS)

CMMDS maintains the authoritative map of all vSAN objects, components, disk groups, and host membership. All hosts must agree on CMMDS state.

#### Layer 4 — Local Log Structured Object Manager (LSOM)

LSOM manages physical read/write operations on local disks. It implements a log-structured write path for maximum SSD/NVMe efficiency.

#### Layer 5 — Physical Disks

The actual cache (SSD/NVMe) and capacity (SSD/HDD/NVMe) devices contributed by each ESXi host.

### vSAN Object & Component Model

All VM data in vSAN is stored as **Objects**. Each VM has the following object types:

- **VM Home Namespace** — stores .vmx, .vmsd, .nvram, logs
- **VMDK Object** — stores virtual disk data (one per disk)
- **VM Swap Object** — memory swap pages during VM power-on
- **Snapshot Delta VMDK** — incremental snapshot changes

Objects are split into **Components** — the smallest storage unit. Components are distributed across hosts per the storage policy. A **Witness Component** holds only metadata for quorum voting.

#### □ NOTE: Component Limit

Each ESXi host supports up to 9,000 components (vSAN 7.0+). Monitor this via vSphere Client □ Cluster □ Monitor □ vSAN □ Virtual Objects.

### vSAN Deployment Models

| Model                    | Description                              | Min Hosts | Use Case                        |
|--------------------------|------------------------------------------|-----------|---------------------------------|
| <b>Standard Cluster</b>  | All hosts in one site contribute storage | 3         | General production workloads    |
| <b>Stretched Cluster</b> | Hosts split across 2 sites + witness     | 2+2+1     | Site-level HA / Zero RPO DR     |
| <b>2-Node Cluster</b>    | 2 data hosts + 1 remote witness          | 2+1       | ROBO / branch office            |
| <b>vSAN Max</b>          | Dedicated storage cluster (storage-only) | 3         | Disaggregated HCI with HCI Mesh |

## CHAPTER 02 · VSAN DISK GROUPS & STORAGE TIERS

### vSAN OSA — Disk Group Architecture

In vSAN Original Storage Architecture (OSA), disks are organized into **Disk Groups** on each ESXi host.

- **Cache Tier:** 1 × SSD or NVMe per disk group — handles all writes (100%) and read caching (for Hybrid). Minimum 10 DWPD endurance required.
- **Capacity Tier:** 1–7 × SSD or HDD per disk group — stores persistent data.
- **Per Host Limit:** up to 5 disk groups = max 5 cache + 35 capacity devices.

| Config Type | Cache Device            | Capacity Device | Dedup/Compress |
|-------------|-------------------------|-----------------|----------------|
| Hybrid      | SSD (R+W cache)         | HDD             | Not supported  |
| All-Flash   | SSD/NVMe (W cache only) | SSD             | Supported      |

### vSAN ESA — Single-Tier Architecture

vSAN Express Storage Architecture (ESA), introduced in vSphere 8.0, eliminates the cache/capacity separation. All-NVMe single-tier pool with higher compression efficiency and RDMA support. Discussed fully in Chapter 11.

#### □ **WARNING: HCL Compliance is Mandatory**

All SSD, NVMe, HDD, and NIC devices must appear on the official vSAN Hardware Compatibility List (HCL/VCG). Non-HCL devices cause health check failures and are unsupported.

### Cache Tier Sizing Guidelines

- **Hybrid clusters:** Size cache SSD at  $\geq 10\%$  of total capacity tier (e.g., 1 TB SSD per 10 TB HDD)
- **All-Flash clusters:** No strict ratio required; cache device handles write buffering only.
- Use **high-endurance SSD** ( $\geq 10$  DWPD) for cache — write-intensive role.
- Replace cache SSDs proactively before end-of-life (monitor via vSAN Health □ Physical Disk).

## CHAPTER 03 · VM STORAGE POLICIES

### What is a VM Storage Policy?

A VM Storage Policy (SPBM — Storage Policy Based Management) is a vSphere object that defines how vSAN provisions and protects VM data. Policies are assigned per-VM or per-VMDK and enforced by the vSAN layer.

### Key Policy Capabilities

| Policy Capability              | Values                 | Meaning                                    |
|--------------------------------|------------------------|--------------------------------------------|
| Failures to Tolerate (FTT)     | 0 / 1 / 2 / 3          | How many simultaneous failures VM survives |
| RAID/Protection Method         | RAID-1, RAID-5, RAID-6 | Mirroring or Erasure Coding                |
| Number of Disk Stripes         | 1-12                   | Stripes per object across capacity devices |
| Object Space Reservation (OSR) | 0-100 %                | 0%=thin, 100%=thick provision              |
| Flash Read Cache Reservation   | 0-100 %                | SSD cache % reserved (Hybrid only)         |
| IOPS Limit                     | 0=unlimited            | Max IOPS cap per VMDK (QoS)                |
| Checksum                       | Enabled/Disabled       | End-to-end data integrity check            |
| Force Provisioning             | Yes/No                 | Provision even if policy cannot be met     |

### RAID Levels — Deep Dive

#### RAID-1 (Mirroring)

- Each write is copied to N+1 hosts simultaneously (synchronous mirroring).
- FTT=1 □ 2 replicas, 3 hosts minimum, **2× storage overhead**.
- FTT=2 □ 3 replicas, 5 hosts minimum, **3× storage overhead**.
- FTT=3 □ 4 replicas, 7 hosts minimum, **4× storage overhead**.
- Best for: latency-sensitive workloads (databases, OLTP).

### RAID-5 (Erasure Coding — 3+1 stripe)

- FTT=1, minimum **4 hosts**, **1.33× storage overhead**.
- Data is split into 3 data stripes + 1 parity stripe across 4 hosts.
- Higher CPU overhead vs RAID-1; best for capacity-critical workloads.

### RAID-6 (Erasure Coding — 4+2 stripe)

- FTT=2, minimum **6 hosts**, **1.5× storage overhead**.
- Double parity — can survive 2 simultaneous failures.
- Maximum space efficiency at FTT=2 vs RAID-1 FTT=2 (3× overhead).

| Protection   | Method | Min Hosts | Overhead | Best For             |
|--------------|--------|-----------|----------|----------------------|
| <b>FTT=0</b> | None   | 1         | 1.0×     | Test/Dev only        |
| <b>FTT=1</b> | RAID-1 | 3         | 2.0×     | Performance-critical |
| <b>FTT=1</b> | RAID-5 | 4         | 1.33×    | Capacity-critical    |
| <b>FTT=2</b> | RAID-1 | 5         | 3.0×     | High availability    |
| <b>FTT=2</b> | RAID-6 | 6         | 1.5×     | Space-efficient HA   |
| <b>FTT=3</b> | RAID-1 | 7         | 4.0×     | Maximum protection   |

#### □ **WARNING: FTT=0 in Production**

Never use FTT=0 for production VMs. A single disk or host failure will cause immediate VM data loss with no recovery path.

## CHAPTER 04 · VSAN NETWORKING

### Network Requirements

| Requirement           | Standard       | Recommended                         |
|-----------------------|----------------|-------------------------------------|
| Bandwidth (Hybrid)    | 1 GbE minimum  | 10 GbE per host                     |
| Bandwidth (All-Flash) | 10 GbE minimum | 25 GbE per host                     |
| Bandwidth (ESA)       | 25 GbE minimum | 100 GbE per host                    |
| MTU                   | 1500 (default) | 9000 (Jumbo Frames) – recommended   |
| Latency (same site)   | Not specified  | < 1 ms RTT between hosts            |
| Protocol              | TCP/IP         | RDMA/RoCEv2 for ESA                 |
| Port used             | TCP 2233 (RDT) | Dedicated VMkernel adapter required |

### VMkernel Configuration

Each ESXi host must have a dedicated VMkernel adapter tagged for vSAN traffic. Best practice is to use a separate physical NIC or dedicated VLAN.

- Tag: **vSAN** service on the VMkernel adapter.
- IP addressing: must be reachable between all hosts in the cluster.
- Verify connectivity: **vmkping -I vmk1 -d -s 8972 <target\_vsan\_ip>**
- -d = do not fragment, -s 8972 = jumbo frame test (9000 MTU – 28 header bytes)

#### □ WARNING: MTU Mismatch Alert

If physical switch ports are set to 1500 MTU and VMkernel is set to 9000 MTU, fragmentation occurs silently, causing severe vSAN performance degradation. Always validate end-to-end MTU consistency before going live.

### Network Partition

A **network partition** occurs when hosts cannot communicate over the vSAN network, creating isolated sub-clusters. CMMDS detects this and evaluates quorum:

- Sub-cluster with >50% of components retains write access.

- Minority sub-cluster goes read-only or VMs are paused to prevent split-brain.
- Resolution: restore network connectivity; CMMDS re-merges sub-clusters.

□ **TIP: Multicast vs Unicast**

vSAN 6.6+ replaced multicast CMMDS discovery with unicast, making vSAN easier to deploy in L3-routed networks. No multicast configuration is needed on modern vSAN deployments.

## CHAPTER 05 · VSAN SECURITY & ENCRYPTION

### Data-at-Rest Encryption

vSAN data-at-rest encryption uses **AES-256** to encrypt all data on capacity disks. Even if a disk is physically stolen, its data is unreadable without the encryption key.

#### □ Key Management Options:

- External KMS (KMIP protocol) — third-party servers like HyTrust, Thales, etc.
- vSphere Native Key Provider (NKP) — built-in since vSphere 7.0 U2, no external server needed.
- Enabling encryption triggers a rolling disk group reformat — plan a maintenance window.
- Data shredding: when encryption is enabled, removing disk groups writes zeros, ensuring data is irrecoverable.

#### □ NOTE: Key Backup is Critical

Always back up the KMS keys or the vSphere Native Key Provider backup bundle. Loss of encryption keys = permanent data loss, even with healthy disks.

### Data-in-Transit Encryption

vSAN data-in-transit encryption uses **AES-256-GCM** to encrypt data as it travels between ESXi hosts over the vSAN network. It protects against network eavesdropping and man-in-the-middle attacks.

- No external KMS required — keys are managed internally by vCenter.
- Can be enabled/disabled independently from data-at-rest encryption.
- Slight CPU overhead (~2-5%) for encryption/decryption operations.

### vSAN RBAC & Access Control

- vSAN uses vSphere RBAC for all management operations.
- Custom roles can restrict specific vSAN permissions (disk management, policy changes).
- All vSAN operations are logged via vCenter audit trail and syslog.
- Integrate with SIEM systems for compliance monitoring (PCI-DSS, HIPAA, etc.).

**▣ TIP: Secure Boot**

Enable UEFI Secure Boot on ESXi hosts to prevent tampered or unauthorized ESXi images from booting. This adds an additional layer of protection for the vSAN storage layer.

## CHAPTER 06 · VSAN SPACE EFFICIENCY

### Deduplication & Compression (OSA All-Flash Only)

Space efficiency technologies reduce the amount of physical storage consumed on capacity devices:

- **Deduplication:** Detects and removes duplicate 4KB data blocks across the entire cluster. Operates in the cache tier before de-staging.
- **Compression:** Compresses unique data blocks before writing to capacity tier. Works on top of deduplication.
- Both are cluster-wide settings — cannot be per-VM.
- Requires **All-Flash** OSA configuration (not supported on Hybrid).
- Enabling triggers a rolling disk group format change (rolling maintenance window required).

| Workload Type           | Typical Dedupe Ratio | Typical Compression | Combined Savings             |
|-------------------------|----------------------|---------------------|------------------------------|
| VDI (linked clones)     | 4:1 to 8:1           | 2:1                 | 60–80% space savings         |
| General VMs             | 1.5:1 to 3:1         | 1.5:1               | 30–50% savings               |
| Databases (Oracle, SQL) | 1.1:1                | 1.2:1               | Minimal savings              |
| Already-compressed data | 1:1                  | 1:1                 | No savings — may slow writes |

#### □ **WARNING: Databases and Deduplication**

Do not enable deduplication/compression for database workloads. Databases write unique, often pre-compressed data, so dedup adds CPU/latency overhead with near-zero space savings.

### Erasure Coding as Space Efficiency

RAID-5/6 erasure coding provides built-in space efficiency as an alternative to mirroring:

- RAID-5 (FTT=1): uses only **1.33×** storage vs 2× for RAID-1 mirroring.
- RAID-6 (FTT=2): uses only **1.5×** storage vs 3× for RAID-1 FTT=2.

- Ideal for large-capacity workloads with moderate performance requirements.

## Slack Space Requirement

vSAN always requires free **slack space** buffer in the datastore:

- Maintain **minimum 20-25% free capacity** at all times.
- Slack space is consumed by: component resync, dedupe staging, snapshot delta growth, metadata overhead, and object placement during host failures.
- Set vSAN capacity alarms at 70%, 80%, and 90% to get proactive warnings.

## CHAPTER 07 · VSAN STRETCHED CLUSTER & HCI MESH

### vSAN Stretched Cluster

A vSAN Stretched Cluster spans two geographically separate data sites with synchronous mirroring between them, providing **zero RPO** site-level protection.

#### Architecture

- **Site A:** Primary data site (Preferred) — active workloads.
- **Site B:** Secondary data site — mirror of Site A data.
- **Site C (Witness):** Third site or cloud hosts only the Witness host.

#### Network Requirements

- **Data Site to Data Site:** ≤5 ms RTT, 10 GbE minimum.
- **Data Sites to Witness:** ≤200 ms RTT, 1 Mbps minimum (per 1000 components).

#### Storage Policy for Stretched Cluster

- **PFTT (Primary Failures to Tolerate):** site-level tolerance (PFTT=1 = one site can fail).
- **SFTT (Secondary Failures to Tolerate):** host-level tolerance within each site.
- Example policy: PFTT=1, SFTT=1, RAID-1 □ requires 2 hosts per site minimum.

#### □ NOTE: Preferred Site

During a network partition where the witness cannot reach both sites, the Preferred (primary) site retains quorum and VM workloads continue. Configure the site with the most critical workloads as Preferred.

### Witness Host & Witness Appliance

- **Witness Host:** ESXi host at third site storing only witness/metadata components.
- **Witness Appliance:** Pre-built OVA deployed as a VM at HQ or cloud — no dedicated hardware needed.
- Witness does NOT store actual VM data — only metadata for quorum voting.
- Witness does NOT require a vSAN production license.

### vSAN HCI Mesh

HCI Mesh allows compute clusters (without local storage) to consume storage from remote vSAN clusters over the network:

- **Compute Cluster:** ESXi hosts with no local vSAN disks — mounts remote vSAN datastore.
- **Storage Cluster:** Dedicated vSAN cluster serving storage to compute clusters.
- vSAN Max: dedicated storage-only cluster model for large-scale disaggregated HCI.

### vSAN Asynchronous Replication

- **RPO:** Minimum 5 minutes (configurable); asynchronous, not synchronous.
- Used for DR scenarios where sites have higher latency (>5 ms RTT).
- Integrates with **VMware Site Recovery Manager (SRM)** for automated DR failover.
- Different from Stretched Cluster: data written to primary, then replicated to secondary.

## CHAPTER 08 · VSAN OPERATIONS & MAINTENANCE

### Maintenance Mode Options

When placing an ESXi host in maintenance mode, vSAN offers three data migration choices:

| Option                      | Data Moved                                   | Speed   | When to Use                           |
|-----------------------------|----------------------------------------------|---------|---------------------------------------|
| <b>Ensure Accessibility</b> | Only components needed for policy compliance | Fastest | Rolling upgrades, short maintenance   |
| <b>Full Data Migration</b>  | ALL components off the host                  | Slowest | Host decommission / permanent removal |
| <b>No Data Migration</b>    | Nothing moved                                | Instant | Emergency only — risk of data loss    |

#### ⚠ WARNING: Resync Before Next Host

NEVER put a second host into maintenance mode until the first host's resync has fully completed. Overlapping maintenance can cause policy violations or data unavailability.

### Component States

| State                | Meaning                                    | Action                              |
|----------------------|--------------------------------------------|-------------------------------------|
| <b>Healthy</b>       | Component accessible and in sync           | None — all good                     |
| <b>Absent</b>        | Host/disk unreachable; within absent timer | Wait — may self-heal on host return |
| <b>Degraded</b>      | Host/disk failed; rebuild timer expired    | Rebuild starts; check hardware      |
| <b>Stale</b>         | Out-of-sync data from host absence         | Resync triggered automatically      |
| <b>Reconfiguring</b> | Policy change in progress                  | Wait for resync to complete         |

### Absent Timer & Rebuild

⚠ **Default Absent Timer:** 60 minutes before vSAN begins rebuilding absent components.

- Purpose: prevents unnecessary rebuild I/O when hosts reboot or have brief network blips.
- After timer expires: vSAN begins rebuilding on remaining healthy hosts.
- Rebuild speed: throttled to protect VM performance; monitored in Resyncing Objects view.

## Disk Replacement Procedure

- Step 1: Verify failure via vSAN Health □ Physical Disk checks.
- Step 2: Use ESXCLI to permanently mark the failed disk (triggers rebuild on other hosts).
- Step 3: Monitor rebuild completion in Resyncing Objects view.
- Step 4: After rebuild: physically replace the disk.
- Step 5: Add new disk to disk group (auto-claim or manual) and verify health.

## CHAPTER 09 · VSAN PERFORMANCE MONITORING

### vSAN Performance Service

The vSAN Performance Service stores historical I/O metrics in a dedicated stats DB object on the vSAN datastore. It provides up to 90 days of history via vSphere Client performance dashboards.

- Enable via: Cluster □ Configure □ vSAN □ Services □ Performance Service.
- Key metrics: IOPS (read/write), Throughput (MB/s), Latency (ms), Outstanding I/O.
- Viewable at: Cluster level, Host level, VM level, and Disk level.

### vSAN Observer (Real-Time)

vSAN Observer is a browser-based real-time monitoring tool launched from the Ruby vSphere Console (RVC) or vSphere Client. It provides live cluster-wide statistics:

- Frontend IOPS / Latency (VM perspective).
- Backend IOPS / Latency (vSAN layer perspective).
- Congestion counters (DOM and LSOM-level).
- Cache hit ratio (OSA Hybrid — critical indicator of working set fit).
- Resync bandwidth consumption.

### vSAN Proactive Tests

Proactive Tests actively generate synthetic workloads to validate cluster readiness:

- **Network Performance Test:** Measures actual bandwidth and latency between all host pairs.
- **Storage Performance Test:** Runs I/O benchmark to validate read/write IOPS and latency.
- Run before production deployments and after any hardware changes.
- Access: vSphere Client □ Cluster □ Monitor □ vSAN □ Proactive Tests.

### Key Performance Indicators to Watch

| Metric                          | Healthy Range     | Action if Exceeded                 |
|---------------------------------|-------------------|------------------------------------|
| <b>Read Latency (frontend)</b>  | < 2 ms            | Check cache hit ratio, congestion  |
| <b>Write Latency (frontend)</b> | < 5 ms            | Check backend write path, resync   |
| <b>Cache Hit Ratio (Hybrid)</b> | > 90%             | Increase cache size or All-Flash   |
| <b>Resync Throughput</b>        | Monitor trend     | Throttle if impacting VM workloads |
| <b>Congestion Score</b>         | 0 (no congestion) | Expand cluster or adjust workloads |
| <b>Component Count/Host</b>     | < 8,500           | Add hosts to distribute components |

#### ▣ TIP: esxtop for vSAN

In esxtop, press "u" for disk device mode. Look for vSAN devices (vsanDevXX) and monitor GAVG (total latency) and KAVG (kernel latency). High KAVG indicates storage stack issues independent of the network.

## CHAPTER 10 · VSAN TROUBLESHOOTING

### vSAN Health Service

The vSAN Health Service performs 200+ automated checks across all cluster components. It is the first tool to use for any vSAN issue.

- Access: vSphere Client □ Cluster □ Monitor □ vSAN □ Skyline Health.
- Categories: Hardware compatibility, Network, Storage device, Data, Performance, Cluster.
- Each failing check links to a KB article with remediation steps.
- Export health data for VMware GSS support bundles.

### Common Issues & Resolutions

| Issue / Symptom                       | Root Cause                                        | Resolution                                                                |
|---------------------------------------|---------------------------------------------------|---------------------------------------------------------------------------|
| <b>MTU mismatch health check fail</b> | VMkernel and switch MTU not matching              | Set all VMkernel adapters and switch ports to MTU 9000 uniformly          |
| <b>Object non-compliant</b>           | FTT cannot be met (host/disk failure)             | Restore failed component; or fix underlying hardware; then Reapply Policy |
| <b>High write latency</b>             | Cache tier full; heavy resync; network congestion | Check resync status; validate network bandwidth; check cache health       |
| <b>Cluster member count mismatch</b>  | Network partition between hosts                   | Verify vSAN VMkernel connectivity via vmkping; check switch config        |
| <b>Object inaccessible</b>            | Too many failures exceed FTT                      | Restore failed hosts/disks immediately; recover from backup if needed     |
| <b>vSAN datastore &gt;80% full</b>    | Insufficient capacity planning                    | Add disk groups or hosts; clean snapshots; enable dedup/compress          |

### Useful Diagnostic Commands

- **esxcli vsan cluster get** — Show cluster UUID, membership, mode.
- **esxcli vsan storage list** — List all vSAN disk groups and drive status.
- **esxcli vsan health cluster list** — Run health checks from CLI.
- **esxcli vsan debug object list** — List all objects and their health.

- ❑ **esxcli vsan debug resync summary** — View resync progress.
- ❑ **vmkping -I vmk1 -d -s 8972 <target\_ip>** — Test vSAN network MTU.
- ❑ **esxcli vsan debug log level set --level verbose** — Enable verbose logging.

❑ **NOTE: vCenter Recovery on vSAN**

If vCenter itself runs on vSAN and becomes inaccessible, use the vCenter bootstrapping procedure: connect directly to ESXi hosts using the vSphere Client host client to power on the vCenter VM via ESXCLI commands before cluster-level management is restored.

## CHAPTER 11 · VSAN ESA — EXPRESS STORAGE ARCHITECTURE

### ESA Overview

vSAN ESA (Express Storage Architecture) is the next-generation vSAN storage architecture introduced in **vSphere 8.0**. It is purpose-built for **NVMe-based All-Flash** hardware and delivers superior performance, compression, and operational simplicity compared to OSA.

### ESA vs OSA Comparison

| Feature          | OSA (Original)                    | ESA (Express)                          |
|------------------|-----------------------------------|----------------------------------------|
| Architecture     | Two-tier (cache + capacity)       | Single-tier (all NVMe)                 |
| Cache concept    | Separate cache SSD per disk group | No separate cache layer                |
| Disk type        | SSD, NVMe, HDD                    | NVMe SSD only (certified)              |
| RAID-5 min hosts | 4 hosts                           | 3 hosts                                |
| Compression      | Post-process, basic               | Inline, superior algorithm (up to 4:1) |
| RDMA support     | Not supported                     | RoCEv2 RDMA for ultra-low latency      |
| Snapshots        | Traditional delta VMDKs           | Copy-on-write with lower overhead      |
| Dedup            | Cluster-wide, All-Flash only      | Enhanced algorithm, better ratio       |
| Management       | Disk group based                  | Pool-based, simplified                 |

### ESA Hardware Requirements

- All storage devices must be **NVMe SSDs** listed on the vSAN ESA HCL.
- Minimum **25 GbE** network per host; 100 GbE recommended for high throughput.
- RDMA/RoCEv2-capable NICs required for ultra-low latency deployments.
- Minimum **3 hosts** for RAID-5 (FTT=1) — down from 4 hosts in OSA.
- ESA clusters cannot mix with OSA disk groups on the same cluster.

**▣ TIP: ESA Key Exam Advantage**

vSAN ESA allows RAID-5 (FTT=1) with only 3 hosts — a key differentiator from OSA which requires 4 hosts for RAID-5. This is a common exam question.

## CHAPTER 12 · VSAN FILE SERVICES & ISCSI TARGET

### vSAN File Service

vSAN File Service enables vSAN to present NFS and SMB file shares directly from the vSAN datastore, turning vSAN into a unified block + file storage platform.

- **Supported Protocols:** NFS v3, NFS v4.1, SMB 2.x/3.x.
- **Architecture:** File Server Agent VMs (FSA VMs) run on each host, managed by vSAN automatically.
- **Use Cases:** Shared home directories, application shared storage, Kubernetes ReadWriteMany (RWX) PVs.
- **Capacity:** Up to 1,024 file shares per cluster.
- Requires **vSAN Advanced** license or higher.

#### □ NOTE: DNS Required for File Service

vSAN File Service requires DNS for share hostnames. Configure forward and reverse DNS entries for the file service IP addresses before creating NFS/SMB shares.

### vSAN iSCSI Target Service

vSAN iSCSI Target Service presents vSAN storage as iSCSI LUNs to physical (bare-metal) servers and non-vSphere hosts.

- **Use Cases:** Physical Oracle/SQL Server, Windows WSFC shared disks, Hadoop nodes, bare-metal Linux.
- Up to 64 iSCSI targets per cluster; multiple LUNs per target.
- Each iSCSI LUN is backed by a vSAN VMDK object with a storage policy.
- Requires **vSAN Advanced** license or higher.

## CHAPTER 13 · VSAN LICENSING & EDITIONS

### vSAN Edition Comparison

| Feature                           | Standard | Advanced | Enterprise |
|-----------------------------------|----------|----------|------------|
| RAID-1/5/6 Policies               | ❑        | ❑        | ❑          |
| Data-at-Rest Encryption           | ❑        | ❑        | ❑          |
| vSAN Health & Performance Service | ❑        | ❑        | ❑          |
| Deduplication & Compression       | ❑        | ❑        | ❑          |
| File Services (NFS/SMB)           | ❑        | ❑        | ❑          |
| iSCSI Target Service              | ❑        | ❑        | ❑          |
| Stretched Cluster                 | ❑        | ❑        | ❑          |
| HCI Mesh / Replication            | ❑        | ❑        | ❑          |
| Data-in-Transit Encryption        | ❑        | ❑        | ❑          |
| vSAN Capacity Advisor             | ❑        | ❑        | ❑          |

### Licensing Model

- ❑ **Per CPU socket:** One vSAN license per physical CPU socket on each ESXi host.
- ❑ **Subscription (vSAN+):** Subscription model with cloud services integration, VMware Aria analytics.
- ❑ **VMware Cloud Foundation (VCF):** vSAN is bundled in VCF — the recommended enterprise deployment model.
- ❑ **Evaluation:** 60-day full Enterprise evaluation license available.
- ❑ **Witness Appliance:** Does NOT require a vSAN production license.

#### ❑ NOTE: License Assignment

Apply vSAN licenses via: vSphere Client ❑ Administration ❑ Licensing ❑ Licenses ❑ Add Key.  
Then: Cluster ❑ Configure ❑ Licensing ❑ Assign License. License expiry blocks new VM provisioning but existing VMs continue.

## CHAPTER 14 · BEST PRACTICES & EXAM TIPS

### Design Best Practices

- **Uniform hardware:** Use identical CPU, RAM, NIC, and storage across all hosts for predictable performance.
- **vSAN Ready Nodes:** Pre-validated server configs from Dell, HPE, Cisco, Lenovo, etc.
- **Dedicated NICs:** Use a dedicated physical NIC for vSAN traffic, separate from VM and management traffic.
- **Jumbo Frames:** Configure MTU 9000 end-to-end (VMkernel + physical switches) for All-Flash clusters.
- **N+1 planning:** Always design for one additional host beyond FTT requirements for maintenance windows.
- **Slack space:** Maintain 20-25% free capacity buffer at all times.

### Critical Exam Formulas & Facts

#### □ NOTE: Must-Memorize for Exam

RAID-1 min hosts =  $(2 \times \text{FTT}) + 1$  | RAID-5 FTT=1 = 4 hosts min, 1.33× overhead | RAID-6 FTT=2 = 6 hosts min, 1.5× overhead | Default absent timer = 60 minutes | Stretched Cluster max site RTT = 5ms | Async replication min RPO = 5 minutes

| Topic                          | Key Fact / Formula                                                     |
|--------------------------------|------------------------------------------------------------------------|
| <b>FTT=1 RAID-1</b>            | Min 3 hosts   2× overhead   Best for performance                       |
| <b>FTT=1 RAID-5</b>            | Min 4 hosts   1.33× overhead   Best for capacity (OSA), 3 hosts in ESA |
| <b>FTT=2 RAID-1</b>            | Min 5 hosts   3× overhead                                              |
| <b>FTT=2 RAID-6</b>            | Min 6 hosts   1.5× overhead   Best space-efficient HA                  |
| <b>FTT=3 RAID-1</b>            | Min 7 hosts   4× overhead   Maximum protection                         |
| <b>Max hosts/cluster</b>       | 64 (standard)   15 per site + witness (stretched)                      |
| <b>Max components/host</b>     | 9,000 (vSAN 7.0+)                                                      |
| <b>Max disk groups/host</b>    | 5 disk groups (OSA)                                                    |
| <b>Max capacity drives/DG</b>  | 7 per disk group (OSA)                                                 |
| <b>Default Absent Timer</b>    | 60 minutes before rebuild starts                                       |
| <b>vSAN Port</b>               | TCP 2233 (RDT — Reliable Datagram Transport)                           |
| <b>vSAN version multicasts</b> | Multicast removed in vSAN 6.6 — unicast only                           |
| <b>Dedup/Compress</b>          | All-Flash OSA only; not supported on Hybrid                            |
| <b>Witness license</b>         | NOT required for Witness Appliance/host                                |
| <b>KMS for encryption</b>      | External KMS or vSphere Native Key Provider (NKP)                      |

## Pre-Exam Checklist

- □ Understand all RAID policy formulas (host count + overhead).
- □ Know all 4 component states: Healthy, Absent, Degraded, Stale.
- □ Know all 3 maintenance mode options and when to use each.
- □ Understand OSA disk group structure (cache + capacity tiers).
- □ Know ESA key advantages: 3-host RAID-5, single-tier, RDMA, better compression.
- □ Know stretched cluster topology: PFTT, SFTT, 5ms RTT limit, witness role.
- □ Understand encryption options: at-rest (KMS/NKP) vs in-transit (no KMS).
- □ Know vSAN Health Service, Observer, and Proactive Tests tools.
- □ Practice vmkping with -d -s 8972 flags for MTU testing.

□ □ Know HCI Mesh: compute-only cluster consuming remote vSAN storage.

□ **TIP: Hands-On Practice**

Complete VMware HOL-2247-01-HCI and HOL-2247-91-HCI hands-on labs before the exam. Practical experience with vSAN Health, storage policies, and maintenance mode operations is tested heavily.

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